Magnetic fields in matter:

All the magnetic phenomena are due to electric charges in motion

magnetic moment per unit volume is magnetization. (After material is kept in magnetic field)

we are not creating magnetic dipoles by passing through a field, we are just aligning them.

⇒ paramagnetism: permanent magnetic moment
(magnetism parallel to $\vec{B}$)
(odd number of electrons)

⇒ diamagnetism: temperory magnetic moment
→ (magnetism opposite to $\vec{B}$)
→ (even number of electrons)

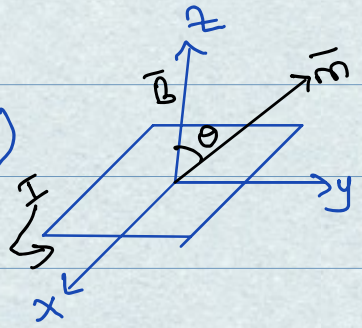
→ present in every material

→ due to orbital motion of electron

⇒ ferromagnetism: neighboring dipoles tend to align along same direction

→ domains of dipoles aligned in same direction

(center of loop is tilted)



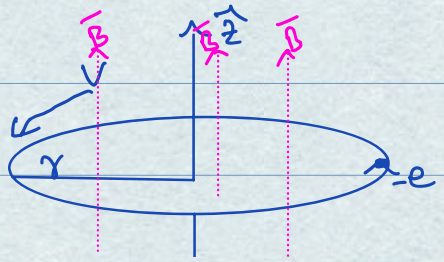
(Net force is zero in uniform field)

Net torque = $\vec{m} \times \vec{B}$ $(\vec{m} = I ab \hat{s})$

→ responsible for paramagnetism (makes $\bar{m}$ align with $\bar{B}$)

Effect of magnetic fields on atomic orbitals

$I = \frac{e}{T}$ $T = \frac{2\pi r}{v}$



$$\Rightarrow I = \frac{e\omega}{2\pi r} \qquad \vec{m} = \frac{e\omega}{2\pi r} \times \pi r^2 (-\hat{z}) \quad (\downarrow \vec{m})$$

$$= -\frac{e\omega r}{2}\hat{z}$$

It's difficult to tilt orbital plane by torque. But speed reduces or increases depending on orientation of B

Before applying $\bar{B}$; $\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{m_e v^2}{r}$

when $\bar{B}$ is applied,

$$\frac{1}{4\pi\varepsilon_0 r^2} e^2 + e\bar{v}B = \frac{m_e \bar{v}^2}{r}$$

$\bar{v}$ is greater than v

If $\bar{v} - v = \Delta v$ is very small, $\Delta v = \frac{Bre}{2m_e}$

dipole moment also increases from m_1 to m_2

So the magnetization is opposite direction to $\bar{B}$

$$\Delta\vec{m} = -\frac{1}{2} e(\Delta v) R\hat{z} = -\frac{e^2 r^2 \vec{B}}{4m_e}$$

For electrons to be accelerated (decelerated some work is to be done. This happens cuz the change in magnetic field produces an electric field which does the work.

The extra dipole moment is responsible for diamagnetism. This is there with all atoms, but much weaker than paramagnetism.

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{R}}{R^2}$$

each volume carries as dipole moment $\vec{M} dV'$

paramagnet is attracted into the field
(field inside > field outside)

diamagnet is repelled (field inside < field outside)
(sometimes zero)
↓
Superconductors

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{M}(\vec{r}')\,\hat{R}}{R^2}\,d\tau' \qquad \vec{\nabla}'(1/R) = \frac{\hat{R}}{R^2}$$

$$= \frac{\mu_0}{4\pi}\int \vec{M}(\vec{r}') \times \left(\vec{\nabla}'(1/R)\right) d\tau'$$

$$\nabla \times fA = f(\nabla \times A) - A \times \nabla f$$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\left[\int \frac{1}{R}\left[\vec{\nabla}' \times \vec{M}(\vec{r}')\right] d\tau' - \vec{\nabla}' \times \left[\frac{\vec{M}(\vec{r}')}{R}\right] d\tau'\right]$$

divergence theorm: $\int \vec{\nabla}\cdot\vec{K}\,d\tau = \oint \vec{K}\cdot d\vec{a}$

put $\vec{K} = \vec{v}\times\vec{c}$ ($\vec{c}$ constant vector)

$$\int \vec{c}\cdot(\vec{\nabla}\times\vec{v})\,d\tau = \int d\vec{a}\cdot(\vec{v}\times\vec{c})$$

$$= -\int \vec{c}\,(\vec{v}\times d\vec{a})$$

$$\Rightarrow \int (\vec{\nabla}\times\vec{v})\,d\tau = -\int (\vec{v}\times d\vec{a})$$

$\nabla\cdot(A\times B) = B\cdot(\nabla\times A) - A\cdot(\nabla\times B)$

$A\cdot(B\times C) = B\cdot(C\times A) = C\cdot(A\times B)$

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi}\left[\int \frac{1}{R}\left[\vec{\nabla}' \times \vec{M}(\vec{r}')\right] d\tau' + \frac{\vec{M}(\vec{r}')}{R}\,d\vec{a}'\right]$$

$$= \frac{\mu_0}{4\pi}\int \frac{1}{R}\left[\vec{\nabla}' \times \vec{m}(\vec{r}')\right] d\tau' + \frac{\mu_0}{4\pi}\oint \frac{1}{R}\left[\vec{m}(\vec{r}') \times d\vec{a}'\right]$$

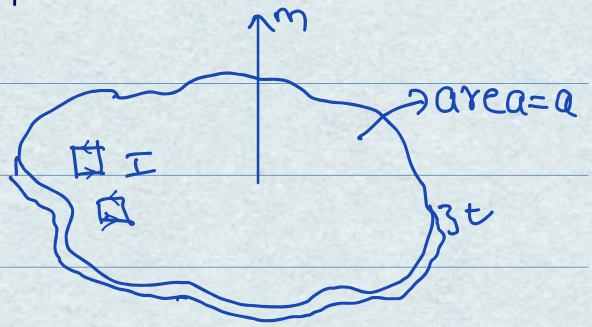
$\vec{J}_b = \vec{\nabla} \times \vec{M}$ → bound volume current

$\vec{K}_b = \vec{M} \times \hat{n}$ → bound surface current

$$\vec{A}_{dip}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_V \frac{\vec{J}_b(\vec{r}')}{R} d\tau' + \frac{\mu_0}{4\pi} \oint_S \frac{\vec{K}_b(\vec{r}')}{R} da'$$

uniformly magnetized: $J_b = \vec{\nabla} \times \vec{m} = 0$

physical interpretation of bound currents



$m = Mat$

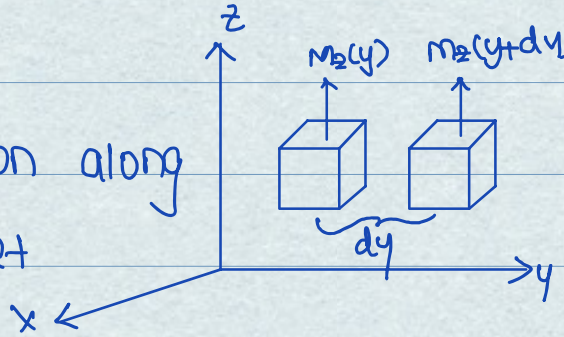
$m = Ia \Rightarrow I = Mt$

$I/t = M = K_b$

$\vec{K}_b = \vec{M} \times \hat{n}$

No current at top and bottom surfaces as m is parallel to n and current vanishes. Current is only along the rim.

Non uniform magnetization along z direction, they'll be net



current in x-direction on surface where they join

$$I_x = [m_z(y+dy) - m_z(y)].dt = \frac{dm_z}{dy}\, dy\, dz$$

Volume current $(J_b)_x = \frac{dm_z}{dy}$

Similarly non-uniform magnetization in y-direction will contribute $(J_b)_x = -\frac{dm_y}{dz} \Rightarrow (J_b)_x = \frac{dm_z}{dy} - \frac{dm_y}{dz}$

All directions: $\vec{J_b} = \vec{\nabla} \times \vec{M}$ and $\vec{\nabla} \cdot \vec{J_b} = 0$

(div of curl is 0)

Auxillary field: current total $= \bar{J} = \bar{J}_b + \bar{J}_f$

$$\frac{1}{\mu_0}(\bar{\nabla} \times \bar{B}) = \bar{J} = \bar{J}_b + \bar{J}_f = (\bar{\nabla} \times \bar{M}) + \bar{J}_f$$

$$\Rightarrow \bar{\nabla} \times \left(\frac{\bar{B}}{\mu_0} - \bar{M}\right) = \bar{J}_f$$

$$\bar{H} = \frac{\bar{B}}{\mu_0} - \bar{M}$$

$\bar{\nabla} \times \bar{H} = \bar{J}_f \longrightarrow$ Ampere's law in terms of H

$\oint \bar{H} \cdot d\bar{l} = (I_f)_{enclosed} \longrightarrow$ Integral form

$\bar{\nabla} \cdot \bar{B} = 0 \Rightarrow \bar{\nabla} \cdot H + \bar{\nabla} \cdot M = 0$

Boundary Conditions:

$\bar{\nabla} \times \bar{H} = \bar{J}_f \qquad \bar{\nabla} \cdot \bar{H} = -\bar{\nabla} \cdot \bar{m}$

$H^{\perp}_{above} - H^{\perp}_{below} = -(M^{\perp}_{above} - M^{\perp}_{below})$

$H''_{above} - H''_{below} = \vec{K}_f \times \hat{n}$

$B^{\perp}_{above} - B^{\perp}_{below} = 0$

$B''_{above} - B''_{below} = \mu_0 (\vec{K}_f \times \hat{n})$

Linear and Non-linear Media:

Linear: $\vec{M} = \mu_0 \chi_m \vec{H}$ ($\chi_m \rightarrow$ magnetic susceptability)

$\bar{B} = \mu_0 (\bar{H} + \bar{M}) = \mu_0 (1 + \chi_m) \bar{H} = \mu \bar{H}$

$\mu = \mu_0 (1 + \chi_m)$ = permeability of material

(for vacuum, $\chi_m = 0$)

$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m$ = relative permeability of the medium

ELECTRODYNAMICS

OHM'S Law: current density $\vec{J}$ is proportional to force per unit charge, $\vec{f}$, the force that makes the charges move to make current flow.

$$\boxed{\vec{J} = \sigma \vec{f}}$$

$\sigma \rightarrow$ conductivity of material

$\rho \rightarrow$ resistivity of material $= \frac{1}{\sigma}$

$$\boxed{\vec{J} = \sigma(\vec{E} + \bar{v} \times \bar{B})}$$

usually the velocity of charges are really small

so $\bar{J} = \sigma \bar{E}$

conductor: $E = 0 \Rightarrow \sigma =$ infinite / very high

resistor: poorly conducting materials, low σ

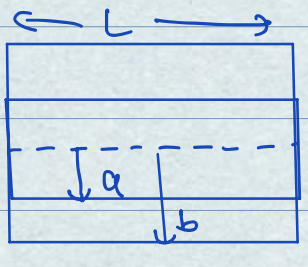
$A \rightarrow$ cross sectional area, $L \rightarrow$ conductor length

$I = JA = \sigma EA = \frac{\sigma A}{L} V \qquad \frac{L}{\sigma A} = R$ (resistance)

$$\bar{E} = \frac{\lambda}{2\pi\varepsilon_0 s} \hat{s}$$

$$I = \int \bar{J} \cdot d\bar{a} = \sigma \int \bar{E} \cdot d\bar{a} = \sigma / \varepsilon_0 \lambda L$$

$$V = -\int \bar{E} \cdot d\bar{l} = \lambda / 2\pi\varepsilon_0 \ln(b/a) \Rightarrow I = 2\pi\sigma L V / \ln(b/a)$$



cylinders

$\Rightarrow \sigma\pi a^2 L / \ln(b/a) \rightarrow$ resistance

$\boxed{\nabla \cdot \vec{E} = \frac{1}{\sigma} \nabla \cdot \vec{J} = 0}$ (for steady uniform current)

$\Rightarrow$ charge density $= 0$

Electromotive force

Force that drives through the circuit

$\vec{F} = \vec{f_s} + \vec{E}$ → electrostatic force

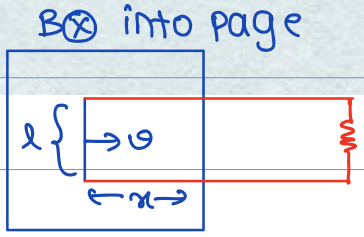
↳ by source

$\mathcal{E} = \oint \vec{F} \cdot d\vec{l} = \oint \vec{f_s} \cdot d\vec{l}$

↳ e.m.f = work done per unit charge

motional e.m.f: when you move a wire through magnetic field (basis of generators)

$\boxed{\mathcal{E} = vBl}$ (clockwise current)



Φ = magnetic flux $= \oint \vec{B} \cdot d\vec{a}$

$= B \cdot l \cdot x$ $\quad \frac{d\Phi}{dt} = Bl\frac{dx}{dt} = -Blv$

$\Rightarrow \boxed{\mathcal{E} = -\frac{d\Phi}{dt}}$ loop can have any shape and can move in any direction

Electromagnetic induction:

$\mathcal{E} = -d\Phi/dt$

$$\Rightarrow \boxed{\oint \bar{E} \cdot d\bar{l} = -\int \frac{d\bar{B}}{dt} \cdot d\bar{a}}$$ Faraday's law

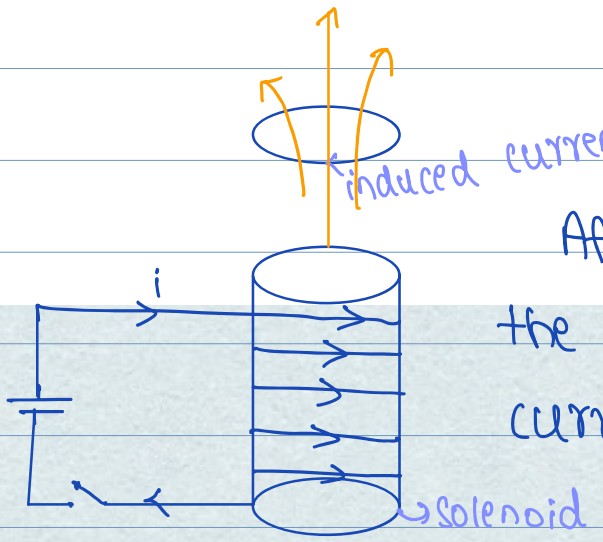
$$\boxed{\bar{\nabla} \times \bar{E} = -\frac{d\vec{B}}{dt}}$$ Differential form of faraday's law

In electrostatics, $\bar{B}$ is constant w.r.t time

↳ no you don't law

Lenz's law: The induced current will flow in such a direction that the flux it produces tends to cancel the change.

The induced emf doesn't depend on the flux but only on the change in flux.



After switch is closed, the ring jumps (opposite currents repel)

Induced electric field:

$$\bar{\nabla} \times \bar{E} = -\frac{d\bar{B}}{dt}$$

↳ induced electric field (faraday field)

changing magnetic field produces an induced e.m.f which produces an induced electric field

$$\bar{\nabla} \cdot \bar{E} = \rho/\varepsilon_0 = 0 \ (\because \rho = 0), \ \bar{\nabla} \cdot \bar{B} = 0$$

↳ faraday field

$$\oint \bar{B} \cdot d\bar{l} = \mu_0 I_{enc}$$

$$\oint \bar{E} \cdot d\bar{l} = -\frac{d\phi}{dt} \longrightarrow \text{Faraday's law in integral form}$$

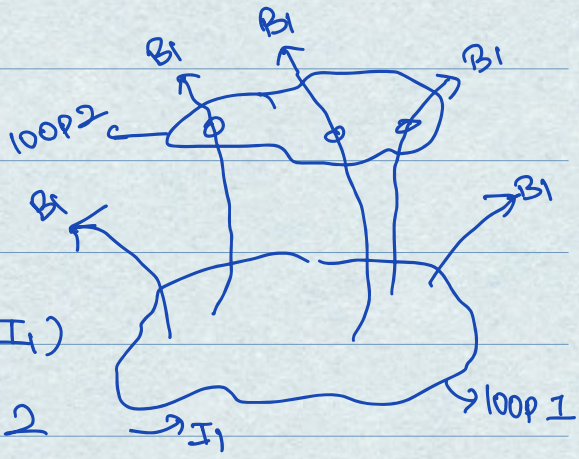
$$\nabla \times \bar{E} = -\frac{d\bar{B}}{dt} = -\frac{d(\bar{\nabla} \times \bar{A})}{dt} = -\bar{\nabla} \times \frac{d\bar{A}}{dt} \Rightarrow \bar{\nabla} \times \left(\bar{E} + \frac{d\bar{A}}{dt}\right) = \bar{0}$$

Since curl is irrotational; this is the gradient of a scalar potential

$$\Rightarrow E + \frac{dA}{dt} = -\nabla\phi \Rightarrow \boxed{E = -\nabla\phi - \frac{dA}{dt}}$$

Stationary field $\Rightarrow \frac{dA}{dt} = 0 \Rightarrow E = -\nabla\phi$ (electrostatics)

"Though using magnetostatic equations in time varying magnetic fields are inappropriate, the error is negligible unless the field changes rapidly. In quasistatic conditions, we can use magnetostatic equations."



Inductance:

$$\bar{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\bar{l}_1 \times \hat{R}}{R^2}$$

(proportional to current I_1)

Φ_2 = flux of B_1 through loop 2

$$\Phi_2 = \int \bar{B}_1 \cdot d\bar{a}_2 = M_{21} I_1$$

M_{21} → mutual inductance of the two loop.

$$\Phi_2 = \oint \bar{B}_1 \cdot d\bar{a}_2 = \oint \bar{A}_1 \cdot d\bar{l}_2 \qquad A_1 = \frac{\mu_0}{4\pi} I_1 \int \frac{d\bar{l}_1}{R}$$

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\bar{l}_1}{R} \right) \cdot d\bar{l}_2 = M_{21} I_1$$

$$\Rightarrow \boxed{M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\bar{l}_1 \cdot d\bar{l}_2}{R}}$$ Newmann's formula

$M_{12} = M_{21} = M$

emf induced is in such a direction as to oppose the change in current, also called back emf

A short solenoid (l, a, n_1) is in the axis of a very long solenoid (b, n_2). I- current flows in small solenoid, to find flux in larger solenoid.

We can assume I flows through larger solenoid and find flux in smaller solenoid.

$$B = \mu_0 n_2 I$$

$$\Phi = \mu_0 n_1 I \times \pi a^2 \times n_1 l = (\mu_0 n_1 n_2 \pi a^2 l) I$$

$$\Rightarrow M_{12} = M_{21} = M = \mu_0 \pi a^2 n_1 n_2 l$$

If current in loop 1 is varied, this induces an emf in loop 2

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M\frac{dI_1}{dt}$$

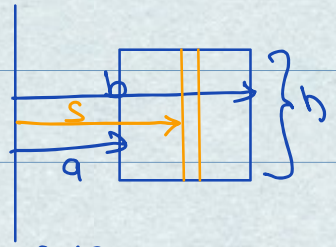
This also induces an emf on loop 1 itself.

Again $\phi = LI$ $\quad L \rightarrow$ Self inductance

$\Rightarrow \mathcal{E} = -L\frac{dI}{dt}$

Self inductance of a toroid with rectangular cross section which carries a total of N turns. (a, b, h)

$B \times 2\pi s = \mu_0 (NI) \Rightarrow B = \frac{\mu_0 NI}{2\pi s}$



$\int \vec{B} \cdot d\vec{a} = \frac{\mu_0 NI}{2\pi} h \int_a^b \frac{ds}{s} = \frac{\mu_0 NIh}{2\pi} \ln\frac{b}{a}$ axis

ϕ (total flux) $= N \times \int B da$

$\Rightarrow L = \frac{\mu_0 N^2 h \ln(b/a)}{2\pi}$

Energy in magnetic fields:

Work done against back emf one trip around the circuit $= -\mathcal{E}$ (unit charge)

charge per unit time $= I$

$\Rightarrow$ Total work per time $\frac{dW}{dt} = -\mathcal{E}I = \frac{LI\,dI}{dt}$

Assuming initial current is 0 to final value I

$\Rightarrow W = \frac{1}{2}LI^2$ (doesn't depend on how long we take to crank up the current)

$\Phi = L \cdot I$

$\phi = \oint \bar{B} \cdot d\bar{a} = \oint \bar{A} \cdot d\bar{l} \Rightarrow LI = \oint \bar{A} \cdot d\bar{l}$

$\Rightarrow W = \frac{1}{2} I \cdot \oint \bar{A} \cdot d\bar{l} = \frac{1}{2} \oint (\bar{A} \cdot \bar{I})\, dl$

$$\boxed{W = \frac{1}{2} \int (\bar{A} \cdot \bar{J})\, d\tau}$$

$\bar{J} = \frac{1}{\mu_0} (\bar{\nabla} \times \bar{B})$

$\Rightarrow W = \frac{1}{2\mu_0} \int \bar{A} \cdot (\bar{\nabla} \times \bar{B})\, d\tau$

$\nabla \cdot (\bar{A} \times \bar{B}) = B \cdot (\nabla \times A) - \bar{A} \cdot (\nabla \times B)$

$\Rightarrow \bar{A} \cdot (\nabla \times B) = \bar{B} \cdot \bar{B} - \nabla \cdot (\bar{A} \times \bar{B})$

$= \frac{1}{2\mu_0} \int_V B^2 d\tau - \frac{1}{2\mu_0} \oint_S (\bar{A} \times \bar{B})\, da$

$$\boxed{W = \frac{1}{2\mu_0} \int_{\text{all space}} B^2 d\tau}$$

$\frac{B^2}{2\mu_0}$ = Energy stored per unit volume

If we find work (W), we can find I by using $L = \frac{2W}{I^2}$

Maxwell's equations:

electrodynamics before maxwell:

$\nabla \cdot \bar{E} = \rho/\varepsilon_0$ Gauss law $\quad \nabla \cdot \bar{B} = 0 \quad \nabla \times \bar{E} = -\frac{d\bar{B}}{dt}$ Faraday's law $\quad \nabla \times \bar{B} = \mu_0 \bar{J}$ Ampere's law

Maxwell fixed Ampere's law

$\nabla \cdot J + \frac{d\rho}{dt} = 0$ (equation of continuity)

(conservation of charge)

$$\nabla \cdot \bar{J} = -\frac{d\rho}{dt} = -\frac{d}{dt}(\varepsilon_0 \nabla \cdot \bar{E})$$

$$\Rightarrow \nabla \cdot \bar{J} = -\nabla \cdot \left(\varepsilon_0 \frac{d\bar{E}}{dt}\right)$$

$$\boxed{\nabla \times \bar{B} = \mu_0 \bar{J} + \mu_0 \varepsilon_0 \frac{d\bar{E}}{dt}}$$

$$\boxed{\varepsilon_0 \frac{d\bar{E}}{dt} = \frac{d\bar{D}}{dt} = \vec{J_d}}$$

displacement current

This displacement current is a current only in a sense that it produces a

magnetic field due to changing electric fields. This displacement current serves to make the total current continuous across discontinuities in conduction current. Displacement current in good conductors is negligible compared to conduction current except at optical frequencies ($\approx 10^{15}$ Hz)

Maxwell's equations in matter:

$(\rho_b)_{polarization} = -\bar{\nabla}\cdot\bar{P}$

$(\bar{J}_b)_{magnetization} = \bar{\nabla}\times\bar{m}$

Total charge density $\rho = \rho_f + \rho_b = \rho_f - \bar{\nabla}\cdot\bar{P}$

Total current density $J = J_f + J_b + J_p = J_f + \bar{\nabla}\times\bar{m} + \frac{d\bar{P}}{dt}$

Substituting in maxwell's law:

$$\bar{\nabla}\cdot\bar{D} = \rho_f \qquad \bar{\nabla}\times\bar{E} = -\frac{d\bar{B}}{dt} \qquad \bar{\nabla}\cdot\bar{B} = 0 \qquad \bar{\nabla}\times\bar{H} = \bar{J}_f + \frac{d\bar{D}}{dt}$$

Displacement current $\vec{J}_D$ can be generalized as

$$\vec{J}_d = \frac{d\bar{D}}{dt}$$

Maxwell's equations in integral form:

$$\oint_S \bar{D} \cdot d\bar{a} = (Q_{free})_{enclosed} \qquad \oint \bar{B} \cdot d\bar{a} = 0$$

Valid over any closed surface

$$\oint_P \bar{E} \cdot d\bar{l} = -\frac{d}{dt} \int_S \bar{B} \cdot d\bar{a} \qquad \oint_P \bar{H} \cdot d\bar{l} = (I_{free})_{enc} + \frac{d}{dt} \oint_S \bar{D} \cdot d\bar{a}$$

Valid for any surface S bounded by closed loop P.

Boundary Conditions:

$$D_1^{\perp} - D_2^{\perp} = \sigma_f \qquad B_1^{\perp} - B_2^{\perp} = 0$$

$$E_1^{\parallel} - E_2^{\parallel} = -\frac{d}{dt} \oint_S \bar{B} \cdot d\bar{a} \qquad H_1^{\parallel} - H_2^{\parallel} = \bar{K}_f \times \hat{n}$$

If there is no free charge of current:

$$\varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp} \qquad E_1^{\parallel} = E_2^{\parallel} \qquad B_1^{\perp} = B_2^{\perp} \qquad \frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}$$

(linear media)

Energy:

Total energy stored in electromagnetic fields is $\boxed{U_{em} = \frac{1}{2}\int \left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right) d\tau}$

$$dW = \bar{F}\cdot d\bar{l} = q(\bar{E} + \bar{v}\times\bar{B})\cdot\bar{v}\,dt$$

$$= q\bar{E}\cdot\bar{v}\,dt$$

$$\rho\bar{v} = \bar{J}, \quad q = \rho\, d\tau$$

$$\Rightarrow dW = \bar{E}\cdot\bar{J}\, dt\, d\tau$$

$$\Rightarrow \frac{dW}{dt} = \int (\bar{E}\cdot\bar{J})\, d\tau$$

$\Rightarrow \bar{E}\cdot\bar{J}$ is power delivered per unit volume

$$\bar{E}\cdot\bar{J} = \frac{1}{\mu_0}\bar{E}\cdot(\bar{\nabla}\times\bar{B}) - \varepsilon_0\bar{E}\frac{\partial\bar{E}}{\partial t}$$

$$\boxed{\bar{\nabla}\cdot(\bar{E}\times\bar{B}) = \bar{B}\cdot(\bar{\nabla}\times\bar{E}) - \bar{E}\cdot(\bar{\nabla}\times\bar{B})}$$

$$\Rightarrow \bar{E}\cdot(\bar{\nabla}\times\bar{B}) = -\bar{\nabla}\cdot(\bar{E}\times\bar{B}) - \bar{B}\cdot\frac{\partial\bar{B}}{\partial t}$$

$$\Rightarrow \boxed{\bar{E}\cdot\bar{J} = -\frac{1}{2}\frac{d}{dt}\left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right) - \frac{1}{\mu_0}\left(\bar{\nabla}\cdot(\bar{E}\times\bar{B})\right)}$$

$$\Rightarrow \boxed{\frac{dW}{dt} = -\frac{d}{dt}\int \frac{1}{2}\left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right) d\tau - \frac{1}{\mu_0}\oint_S (\bar{E}\times\bar{B})\cdot d\bar{a}}$$

Poynting's theorm | Work energy theorm in electrodynamics

The energy per unit time per unit area transported by the fields is called poynting vector $\boxed{\bar{S} = \frac{1}{\mu_0}(\bar{E} \times \bar{B})}$

$\bar{S}.d\bar{a}$ is the energy flux

$$\frac{dW}{dt} = -\frac{dU_{enc}}{dt} - \oint_S \bar{S}.d\bar{a}$$

Total work done on charges will increase their mechanical energy. If we denote u_{mech} as mechanical energy density:

$$\frac{dW}{dt} = \frac{d}{dt}\int u_{mech}\, d\tau$$

$$\Rightarrow \boxed{\frac{d}{dt}\int (u_{mech} + u_{em})\, d\tau = -\oint_S \bar{S}.d\bar{a}}$$

Electromagnetic Waves:

In terms of free charges / currents:

$$\nabla\cdot\vec{D}=\rho_f \quad \nabla\cdot\vec{B}=0 \quad \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \quad \nabla\times\vec{H}=\vec{J}_f+\frac{\partial\vec{D}}{\partial t}$$

When there're no free charges / currents:

$$\nabla\cdot\vec{E}=0 \quad \nabla\cdot\vec{B}=0 \quad \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \quad \nabla\times\vec{B}=\mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}$$

$$\Box^2=\nabla^2-\mu_0\varepsilon_0\frac{\partial^2}{\partial t^2} \rightarrow \text{wave operator / dealembert operator}$$

wave equation

$$\Rightarrow \Box^2 E=0 \Rightarrow \boxed{\nabla^2 E=\mu_0\varepsilon_0\frac{\partial^2 E}{\partial t}}$$

$$\Rightarrow \Box^2 B=0 \Rightarrow \boxed{\nabla^2 B=\mu_0\varepsilon_0\frac{\partial^2 B}{\partial t}}$$

(using curl of last eqns)

$$\vec{E}(\vec{r},t)=|E_0|\hat{n}\cos(\vec{k}\cdot\vec{r}-\omega t+\delta)$$

$\hat{n}\Rightarrow$ direction of electric field also called polarization of plane waves

$\delta\rightarrow$ arbitrary phase

$$\vec{B}(\vec{r},t)=B_0\hat{n}'\cos(\vec{k}\cdot\vec{r}-\omega t+\delta')$$

Plane E.M waves:

wavefront $\rightarrow$ surface of equal phase

$\Rightarrow \bar{K}\cdot\bar{r} - \omega t + \delta = \text{const} \Leftrightarrow \bar{K}\cdot\bar{r} - \omega t = \text{const}$

At given time; $\bar{K}\cdot\bar{r} = \text{const}$

$$\vec{E}(\vec{r},t) = Re\left(E_0 \hat{n} e^{i(\bar{K}\cdot\bar{r}-\omega t)}\right) \qquad E_0 = |E_0| e^{i\delta}$$

$$\vec{B}(\vec{r},t) = Re\left(B_0 \hat{n}' e^{i(\bar{K}\cdot\bar{r}-\omega t)}\right) \qquad B_0 = |B_0| e^{i\delta_1}$$

$$\Rightarrow \vec{E}(\bar{r},t) = E_0 \hat{n} e^{i(\bar{K}\cdot\bar{r}-\omega t)} \; ; \; \vec{B}(\bar{r},t) = B_0 \hat{n}' e^{i(\bar{K}\cdot\bar{r}-\omega t)}$$

$$\frac{d}{dt} e^{-i\omega t} = -i\omega e^{i\omega t}$$

$$\nabla e^{i\bar{K}\cdot\bar{r}} = i\vec{K} e^{i\bar{K}\cdot\bar{r}}$$

$$\Rightarrow \nabla\cdot\left(e^{i\bar{K}\cdot\bar{r}}\hat{n}\right) = (i\bar{K}\cdot\hat{n}) e^{i\bar{K}\cdot\bar{r}}$$

we can use replacements like:

$$\nabla = i\bar{K} \text{ and } \frac{d}{dt} = -i\omega$$

in wave equations;

$$\Rightarrow \mu_0\varepsilon_0\omega^2 - K^2 = 0 \Rightarrow \omega = cK \qquad c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

$$\Rightarrow \omega/K = v_\lambda = c$$

1. EM waves propogate in free space with velocity c.

put $\bar{E}, \bar{B}$ in divergence equations $\Rightarrow$

$\vec{K} \cdot \hat{n} = 0 \; ; \; \vec{K} \cdot \hat{n}' = 0$

$\vec{K} \rightarrow$ wave propogation direction

2. EM waves are transverse waves: Both electric field propogation and magnetic field propogation are normal to one-another

$\Rightarrow \hat{n} \cdot \hat{n}' = 0$

$E_0 (\vec{K} \times \hat{n}) - \omega B_0 \hat{n}' = 0 \Rightarrow \frac{E_0}{c} (\hat{k} \times \hat{n}) = B_0 \hat{n}'$

$B_0 (\vec{K} \times \hat{n}') + \omega \mu_0 \varepsilon_0 E_0 \hat{n} = 0 \Rightarrow B_0 c (\hat{n}' \times \hat{K}) = E_0 \hat{n}$

3. E, B, K Forms a right handed triad

4. Phases of electric and magnetic is same

5. Amplitudes are related by $B_0 = \frac{E_0}{c}$

Summarizing:

- $\bar{E}(\bar{r},t) = |E_0|\hat{n}\cos(\bar{k}\cdot\bar{r}-\omega t+\delta)$
- $\bar{B}(\bar{r},t) = |E_0|\frac{(\hat{k}\times\hat{n})}{c}\cos(\bar{k}\cdot\bar{r}-\omega t+\delta)$
- $\hat{k}\cdot\hat{n}=0 \quad \omega = ck$
- $k \rightarrow$ gives direction of propogation and wavelength $\lambda = 2\pi/k$
- The polarization $\hat{n}$ which gives direction of electric field
- The complex number E_0, which specifies the amplitude and phase of Electric field.

Energy and momentum:

$$u = \frac{1}{2}\int\left(\varepsilon_0 E^2 + \frac{B^2}{\mu_0}\right)d\tau = \varepsilon_0 E^2 = \varepsilon_0 E_0^2 \cos^2(\bar{k}\cdot\bar{r}-\omega t+\delta)$$

contributions from electric and magnetic field are equal

$$\vec{S} = \frac{1}{\mu_0}(\bar{E}\times\bar{B}) = c\varepsilon_0 E_0^2 \cos^2(\bar{k}\cdot\bar{r}-\omega t+\delta) = c\cdot u\cdot\hat{e_z}$$

In time dt, length cdt passes through area A, with energy $Acdt\cdot u$. Energy per unit area per

unit time = c.u (energy flux density → $\vec{S}$)

E.m field carries momentum too

$\vec{g} = \frac{\vec{S}}{c^2} = \frac{u}{c}\cdot\hat{e}_z$ $\hat{z}$: ($\vec{E}$ along $\hat{x}$, $\vec{B}$ along $\hat{y}$)

g → momentum flux density

$\langle u \rangle = \frac{1}{2}\varepsilon_0 E_0^2$

$\langle \vec{S} \rangle = \frac{1}{2} c \varepsilon_0 E_0^2 \hat{e}_z = I$ (intensity)

$\langle \vec{g} \rangle = \frac{1}{2c}\varepsilon_0 E_0^2 \hat{e}_z$

Plane waves in mater

Source free region:

$\bar{\nabla}\cdot\bar{D} = 0$; $\bar{\nabla}\cdot\bar{B} = 0$; $\bar{\nabla}\times\bar{E} = -\frac{\partial \bar{B}}{\partial t}$; $\bar{\nabla}\times\bar{B} = \frac{\partial \bar{D}}{\partial t}$

medium is linear: $\bar{D} = \varepsilon\bar{E}$; $\bar{H} = \frac{\bar{B}}{\mu}$

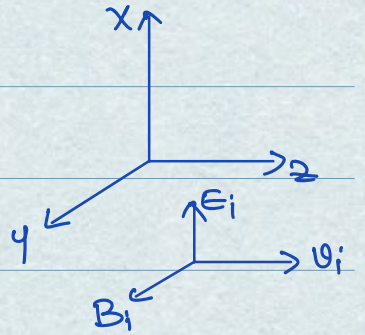
$\bar{\nabla}\cdot\bar{E} = 0$ $\bar{\nabla}\cdot\bar{B} = 0$ $\bar{\nabla}\times\bar{E} = -\frac{\partial \bar{B}}{\partial t}$ $\bar{\nabla}\times\bar{B} = \mu\varepsilon_0\frac{\partial \bar{E}}{\partial t}$

refractive index $n = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} \cong \sqrt{\varepsilon_r}$

Reflection, Refraction of Plane waves at Normal incidence:

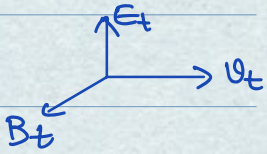
$$E_I(z,t) = E_{0,I}\, e^{i(k_1 z - \omega t)}\,\hat{x}$$

$$B_I(z,t) = \frac{E_{0,I}}{v_1}\, e^{i(k_1 z - \omega t)}\,\hat{y}$$



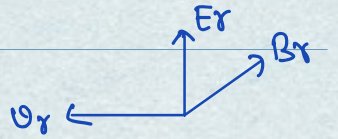
$$E_R(z,t) = E_{0,R}\, e^{i(k_1 z - \omega t)}\,\hat{x}$$

$$B_R(z,t) = \frac{E_{0,R}}{v_1}\, e^{i(k_1 z - \omega t)}\,(-\hat{y})$$



$$E_T(z,t) = E_{0,T}\, e^{i(k_2 z - \omega t)}\,\hat{x}$$

$$B_T(z,t) = \frac{E_{0,T}}{v_2}\, e^{i(k_2 z - \omega t)}\,\hat{y}$$



$$k_1 v_1 = k_2 v_2$$

$$\varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp} \qquad E_1^{\parallel} = E_2^{\parallel} \qquad B_1^{\perp} = B_2^{\perp} \qquad \frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}$$

→ opposite direction

$$E_{0,R} + E_{0,I} = E_{0,T}$$

$$E_{0,I} - E_{0,R} = \beta E_{0,T} \qquad \beta = \frac{\mu_1 v_1}{\mu_2 v_2}$$

$$\frac{E_{0,R}}{E_{0,I}} = \frac{1-\beta}{1+\beta} = \frac{v_2 - v_1}{v_2 + v_1} = \frac{n_1 - n_2}{n_1 + n_2}$$

(assuming $\mu_1 \approx \mu_2$)

$$\frac{E_{0,T}}{E_{0,I}} = \frac{2\beta}{1+\beta} = \frac{2v_2}{v_1 + v_2} = \frac{2n_1}{n_1 + n_2}$$

$$\frac{1}{2} v \varepsilon_0 E_0^2 = I$$

reflection coefficient $R = \frac{I_R}{I_I} = \left(\frac{E_{0,R}}{E_{0,I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$

transmission coefficient $T = \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{E_{0,T}}{E_{0,I}}\right)^2 = \frac{4 n_1 n_2}{(n_1 + n_2)^2}$

$$R + T = 1$$